

08 07 2026

Day Month Year

Calculation:

~~wind~~ q_z = wind velocity pressure

$$q_z = 0.613 \cdot K_z \cdot K_{zt} \cdot K_d \cdot V^2$$

q_z = velocity pressure

K_z = Exposure Coefficient

K_d = wind directionality factor

V = basic wind speed

K_{zt} = Topographic Factor

after field work and research, the conditions where guyed tower ~~unit~~ was installed, ~~the~~ values of

$$K_d = 0.85, K_{zt} = 1$$

While if using ASCE 7, there is not a single fixed value.

~~As suitable~~

for exposure D (as near sea shore) :

$$\alpha = 11.5$$

$$Z_g = 213$$

$$K_z = 2.01 \left(\frac{z}{Z_g} \right)^{2/\alpha}$$

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z_g = Height Gradient

z = height (mean)

α = exposure $D = 11.5$

Mean Height of each section in metres

A

B

C

$$h = 15 = \frac{4.6}{2.13} m$$

$$h = 30 = 9.144 m$$

$$C = 50 = 15.24 m$$

E

D

C

B

A

D

E

$$d = 70 = 21.336 m$$

$$e = 27.432 m$$

K_z for each section

A

$$K_z = 2.01 \left(\frac{4.6}{2.13} \right)^{2/11.5}$$

$$K_z = 0.976/0.41 = 1.0316$$

B

$$K_z = 1.1625$$

E

$$K_z = 1.4073$$

C

$$K_z = 1.2706$$

D

$$K_z = 1.3471$$



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$$V = 25 \text{ km/h} = 6.944 \text{ m/s}$$

Wind Velocity Pressure

$$q_A = 0.613 \cdot (1.0316) \cdot (1) \cdot (0.85) \cdot (6.944)^2$$

$$q_A = 24.1296 \text{ Pa}$$

$$q_B = 0.613 \cdot (1.1625) \cdot (1) \cdot (0.85) \cdot (6.944)^2$$

$$q_B = 29.2073 \text{ Pa}$$

$$q_C = 0.613 \cdot (1.2706) \cdot (1) \cdot (0.85) \cdot (6.944)^2$$

$$q_C = 31.9233 \text{ Pa}$$

$$q_d = 0.613 \cdot (1.3471) \cdot (1) \cdot (0.85) \cdot (6.944)^2$$

$$q_d = 33.8453 \text{ Pa}$$



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$$q_e = 0.613 (1.4073) \cdot (11) (0.85) \times (6.944)^2$$

$$q_e = 35.3578 \text{ Pa}$$

Wind force on Mast

$$F = q_z G C_f A_f$$

$A = \text{Projected Area} = D \times H$

$C_f = \text{force Coefficient} = 0.9 (\text{Circular}) / 0.8 / 1.2$

$G = 0.85$

$$F_A = (25.9185) (0.85) (0.9) (5.5717)$$

$$F_A = 110.47 \text{ N}$$

$$F_A = 105.53 \text{ N}$$

$$F_A = 110.47$$

$$A = 0.914 \times 6.096 =$$

$$F_B = (29.2073) (0.85) (0.9) (5.5717)$$

$$F_B = 127.74 \text{ N}$$

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$$F_c = (31.9233)(0.85)(0.9)(5.5717)$$
$$F_c = 136.06 \text{ N}$$

$$F_d = 33.8453(0.85)(0.9)(5.5717)$$
$$F_d = 144.25 \text{ N}$$

$$A = 6.096 \times 1.638 = 3.72$$

$$F_e = 35.3578(0.85)(0.9)(3.72)$$
$$F_e = 100.62 \text{ N}$$